A-Level Physics: Synoptic Question on Gamma Knife Radiosurgery

Synoptic Question: Physics of the Gamma Knife

The Gamma Knife is a medical device used to treat brain tumors non-invasively. It uses approximately 201 sources of Cobalt-60 ($^{60}_{27}$Co) arranged in a hemispherical array. These sources aim narrow beams of gamma radiation at a single focal point (the "isocenter"), ensuring high dosage to the tumor while sparing surrounding healthy tissue.

Part A: Nuclear Decay and Activity

Cobalt-60 decays into Nickel-60 ($^{60}_{28}$Ni) via beta-minus decay, followed by the emission of two distinct gamma-ray photons.

1. Write the nuclear equation for the decay of $^{60}_{27}$Co to $^{60}_{28}$Ni, including the decay particles.
2. The half-life of Cobalt-60 is 5.27 years.
   - Calculate the decay constant λ in s^{-1}.
   - A new Gamma Knife unit contains 201 sources, each with an initial activity of 1.1×10^{12} Bq. Calculate the total activity of the unit after 10 years of operation.

Part B: Photon Energy and Intensity

Each decay of Cobalt-60 produces two gamma photons with energies of 1.17 MeV and 1.33 MeV.

3. Calculate the frequency of the 1.33 MeV photon. (1 eV $= 1.60 \times 10^{-19}$ J, $h = 6.63 \times 10^{-34}$ J s).
4. The beams are collimated into narrow pencils of radiation. If a single beam has a cross-sectional area of 1.2×10^{-5} m^2 at the isocenter, calculate the photon flux (number of photons per second per unit area) for one source, assuming 100% of emitted photons reach the isocenter.

Part C: Dosimetry and Thermal Effects

The primary goal is to deliver a specific "Absorbed Dose," measured in Grays (Gy), where 1 Gy = 1 J kg^{-1}.

5. A tumor of mass 15 g is located at the isocenter. The treatment plan requires a dose of 25 Gy.
   - Calculate the total energy that must be absorbed by the tumor.
   - Assuming the average energy of the photons is 1.25 MeV and only 40% of the incident photon energy is actually absorbed by the tumor tissue, calculate the total number of photons required to deliver this dose.
6. Explain, with reference to the "Inverse Square Law," why the hemispherical arrangement of many low-intensity sources is safer for the patient than using a single high-intensity beam directed from one side.

Suggested Answers and Mark Scheme

Part A: Nuclear Decay

1. **(2 marks)** ${}^{60}_{27}\text{Co} \rightarrow {}^{60}_{28}\text{Ni} + {}^{0}_{-1}\beta + \bar{\nu}_e + \gamma$
   (Must include electron/beta particle and anti-neutrino).

2. **(4 marks)** $\lambda = \frac{\ln 2}{T_{1/2}} = \frac{0.693}{5.27 \times 365.25 \times 24 \times 3600} \approx 4.17 \times 10^{-9}\ \text{s}^{-1}$.
   $A = A_0 e^{-\lambda t} = (201 \times 1.1 \times 10^{12}) \times e^{-(\frac{\ln 2}{5.27}) \times 10}$
   $A = (2.211 \times 10^{14}) \times 0.268 \approx 5.93 \times 10^{13}$ Bq.

Part B: Photon Energy and Intensity

3. **(2 marks)** $E = 1.33 \times 10^6 \times 1.60 \times 10^{-19} = 2.128 \times 10^{-13}$ J.
   $f = E/h = \frac{2.128 \times 10^{-13}}{6.63 \times 10^{-34}} \approx 3.21 \times 10^{20}$ Hz.

4. **(3 marks)** Each decay produces 2 photons. For 1 source (1.1×10^{12} decays/s):
   Photons per second $= 2.2 \times 10^{12}\ \text{s}^{-1}$.
   Flux $= \frac{\text{Photons per second}}{\text{Area}} = \frac{2.2 \times 10^{12}}{1.2 \times 10^{-5}} \approx 1.83 \times 10^{17}$ photons $\text{m}^{-2}\ \text{s}^{-1}$.

Part C: Dosimetry and Thermal Effects

5. **(3 marks)** Energy $E = \text{Dose} \times \text{mass} = 25 \times 0.015 = 0.375$ J.
   Energy per photon $= 1.25 \times 10^6 \times 1.6 \times 10^{-19} = 2.0 \times 10^{-13}$ J.
   Total photons needed $= \frac{0.375}{0.40 \times 2.0 \times 10^{-13}} \approx 4.69 \times 10^{12}$ photons.

6. **(3 marks)** The intensity of a single beam follows $I = P/A$. If a single high-intensity beam is used, the healthy tissue along the entry path receives a very high dose ($I \propto 1/r^2$). By using 201 separate low-intensity beams, the "entry dose" at any single point on the scalp is 1/201 of the total dose. These beams only overlap at the isocenter, ensuring the high intensity is localized strictly to the tumor.

Cambridge International AS & A Level Physics (9702)
Paper 4 Variant 2 (Extended Difficulty)
Topic Focus: Gamma Knife Radiosurgery (Synoptic)

Instructions (CIE P4 style)

- Answer all questions.
- Show your working and appropriate significant figures.
- Use: $c = 3.00 \times 10^{8}\,\mathrm{m\,s^{-1}}$, $h = 6.63 \times 10^{-34}\,\mathrm{J\,s}$, $e = 1.60 \times 10^{-19}\,\mathrm{C}$, $N_A = 6.02 \times 10^{23}\,\mathrm{mol^{-1}}$.

Total: 30 marks

Question 1: Physics of Gamma Knife Radiosurgery (Variant 2)

Comprehension passage

A neurosurgical unit uses a Gamma Knife system containing **201 cobalt-60 sources** mounted on a hemispherical shell. The distance from each source to the isocentre is **0.35 m**. Each source currently has activity 8.5×10^{11} Bq. Cobalt-60 has half-life **5.27 years** and emits two gamma photons per decay, of energies **1.17 MeV** and **1.33 MeV**. A planned treatment session lasts **32.0 min**.

The tumour at the isocentre has mass **18.0 g**. During dose planning, physicists estimate that only **32%** of incident gamma energy is absorbed by tumour tissue.

Use data from the comprehension passage above where required.

1. Write the full nuclear decay equation for cobalt-60 and explain why two gamma photons are emitted. [3]

2. Calculate the decay constant λ of cobalt-60 in $\mathrm{s^{-1}}$. [2]

3. Determine the total number of decays from all sources during one treatment session. [3]

4. Calculate the total gamma energy emitted by the unit during one treatment session. [4]

5. Using the stated energy-absorption fraction, find the absorbed dose (in Gy) delivered to the tumour in this session. [4]

6. A clinician proposes replacing the 201-source arrangement with a single source delivering the same tumour dose from one direction.

Discuss the physics reasons this is not acceptable. Your answer should include:

- inverse-square law and beam geometry,
- dose distribution in healthy tissue,
- attenuation and scattering,
- at least one practical safety/engineering consideration.

[6]

7. Open-ended design task.

For the same tumour, propose two physics-based modifications to improve treatment quality while keeping risk to surrounding tissue low. For each modification:

- state the physical principle,
- explain expected benefit,
- identify one limitation or trade-off.

[8]

Sample Solution and Mark Scheme

9702 Paper 4 Variant 2 only

1. Nuclear equation and gamma emission [3]

$^{60}_{27}\text{Co} \rightarrow {}^{60}_{28}\text{Ni} + {}^{0}_{-1}\beta + \bar{\nu}_e + \gamma_1 + \gamma_2$.

M1: correct nuclide numbers and beta-minus/antineutrino shown.
A1: daughter nucleus is $^{60}_{28}\text{Ni}$.
B1: explanation that ^{60}Ni is formed excited and de-excites via two discrete gamma transitions.

2. Decay constant [2]

$$\lambda = \frac{\ln 2}{t_{1/2}} = \frac{0.693}{5.27 \times 365.25 \times 24 \times 3600} = 4.17 \times 10^{-9}\,\text{s}^{-1}.$$

M1 substitution of half-life in seconds.
A1 correct value to 3 s.f.

3. Total decays in 32.0 min [3]

$$A_{\text{total}} = 201 \times 8.5 \times 10^{11} = 1.7085 \times 10^{14}\,\text{s}^{-1}.$$

$$t = 32.0 \times 60 = 1920\,\text{s}, \quad N = A_{\text{total}} t = 3.28 \times 10^{17}\,\text{decays}.$$

M1 total activity.
M1 use $N = At$.
A1 correct final decays.

4. Total gamma energy emitted [4]

Energy per decay:

$$E_d = (1.17 + 1.33)\,\text{MeV} = 2.50\,\text{MeV} = 2.50 \times 10^6 \times 1.60 \times 10^{-19} = 4.00 \times 10^{-13}\,\text{J}.$$

Total emitted energy:

$$E_{\text{emit}} = N E_d = (3.28 \times 10^{17})(4.00 \times 10^{-13}) = 1.31 \times 10^5\,\text{J}.$$

M1 converts MeV to J.
M1 finds energy per decay.
M1 multiplies by total decays.
A1 correct total energy.

5. Absorbed dose [4]

Absorbed energy:

$$E_{\text{abs}} = 0.32\,E_{\text{emit}} = 0.32(1.31 \times 10^5) = 4.20 \times 10^4\,\text{J}.$$

Tumour mass $m = 18.0\,\text{g} = 1.80 \times 10^{-2}\,\text{kg}$.

$$D = \frac{E_{\text{abs}}}{m} = \frac{4.20 \times 10^4}{1.80 \times 10^{-2}} = 2.33 \times 10^6\,\text{Gy}.$$

M1 applies 32% absorption factor.
M1 converts mass to kg.

M1 uses $D = E/m$.
A1 correct Gy value (accept 2.3×10^6 Gy).

6. Why single high-intensity source is unacceptable [6]

Indicative points (any 6 valid physics points):

- Inverse-square law means very high intensity near entrance path for a single beam.
- Multi-beam arrangement spreads entry dose over many different directions.
- Only at isocentre do beams superpose to produce high tumour dose.
- Single-beam approach overdoses healthy tissue before and after tumour.
- Attenuation/scattering in skull/tissue broadens dose profile and increases off-target dose.
- Higher local heating/radiation damage risk to critical structures.
- Collimation and patient immobilisation tolerances become more severe for one beam.
- Redundancy/reliability: failure of one source in array has small effect; failure in single-source system is critical.

Use level-based marking if required: coherent chain with equations/physics terms for top marks.

7. Open-ended design improvements [8]

Award up to 4 marks per modification (max 8):

- M1: states valid physics-based modification.
- M1: identifies correct underlying principle.
- A1: clear explanation of tumour-control or tissue-sparing benefit.
- B1: realistic limitation/trade-off.

Example high-scoring response (two modifications):

1. **Adaptive collimator aperture:** Use smaller apertures near organs-at-risk. Principle: geometric beam shaping and fluence control. Benefit: steeper dose gradient around tumour edge. Trade-off: longer treatment time and stricter alignment uncertainty.
2. **Image-guided fractionation:** Deliver dose over multiple sessions with imaging before each session. Principle: repair kinetics and reduced normal-tissue complication probability for sub-lethal damage. Benefit: same total biological effect on tumour with reduced peak normal-tissue dose. Trade-off: greater logistical complexity and possible tumour repopulation between fractions.

Total = 30 marks